



Computational fluid dynamics (CFD) simulation modeling for the cultivation of microalgal monoculture in axenic enclosed bubble column photobioreactor (BCPBR)

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ABSTRACT

Researchers are more concerned with axenic-enclosed PBRs, where there is less or no chance of contamination during the production of biochemical and highly valuable metabolites, and monocultures of microalgae are being grown more frequently. It is a closed, manufactured vessel that aids in the photosynthesis of microalgal cells using artificial light or sunlight as the energy source. In this study, the bubble column PBR (BCPBR) was selected because it possessed some advantages over other PBRs for the growth of *Chlorella vulgaris*. The BCPBR system prevents contact between the enclosed microalgal cells and the environment, allowing the culturing of microalgal species that are difficult to grow in open pond systems. To compare BCPBR performance quantitatively, the efficient mixing expected in BCPBR, as discussed in the literature, was applied to the CFD model. The experimental results observed during the cultivation of *C. vulgaris* with restaurant wastewater (RWW) in BCPBR clearly showed better mixing, high growth, and improved treatment efficiency. CFD analysis was conducted on the evolution of bubbles in the BCPBR. The Pressure-Implicit with Splitting of Operators (PISO) pressure correction method is used for velocity and pressure coupling. A geo-reconstruct approach is used to construct the interface, and a second-order upwind calculation technique is used to determine the flow parameters. Therefore, CFD simulation in this study will contribute to the following aspects: (i) the volume fraction contours and velocity contours are going to validate the experimental study as the homogenous mixing favors the growth and productivity, (ii) To study how the size of the nozzle and inlet velocity affect the turbulence generated by bubbles in a BCPBR to identify the optimal nozzle size and velocity for the required turbulence.

1. Introduction

The rapid increase in population significantly affects the available resources such as food, energy, water consumption, and pollution, which all raise the climatic change problem. This creates a need to rely on eco-friendly sources that can be used as raw materials for several industries and reduce the use of edible plant crops. Algae are considered a valuable source of biomass that can be included in a number of

industries, such as pharmaceutical, cosmetics, food industries, biofuel production, nanoparticle biosynthesis, and wastewater treatment. Microalgae can be cultivated in large-scale photobioreactors to obtain a large amount of biomass that contains valuable compounds such as pigments, carotins, phenols, polysaccharides, oils, proteins, and vitamins. Closed photobioreactors (PBRs) are controlled cultivation systems of microalgae that are protected from direct environmental contact and prohibit contamination and water evaporation [1,2]. The most popular

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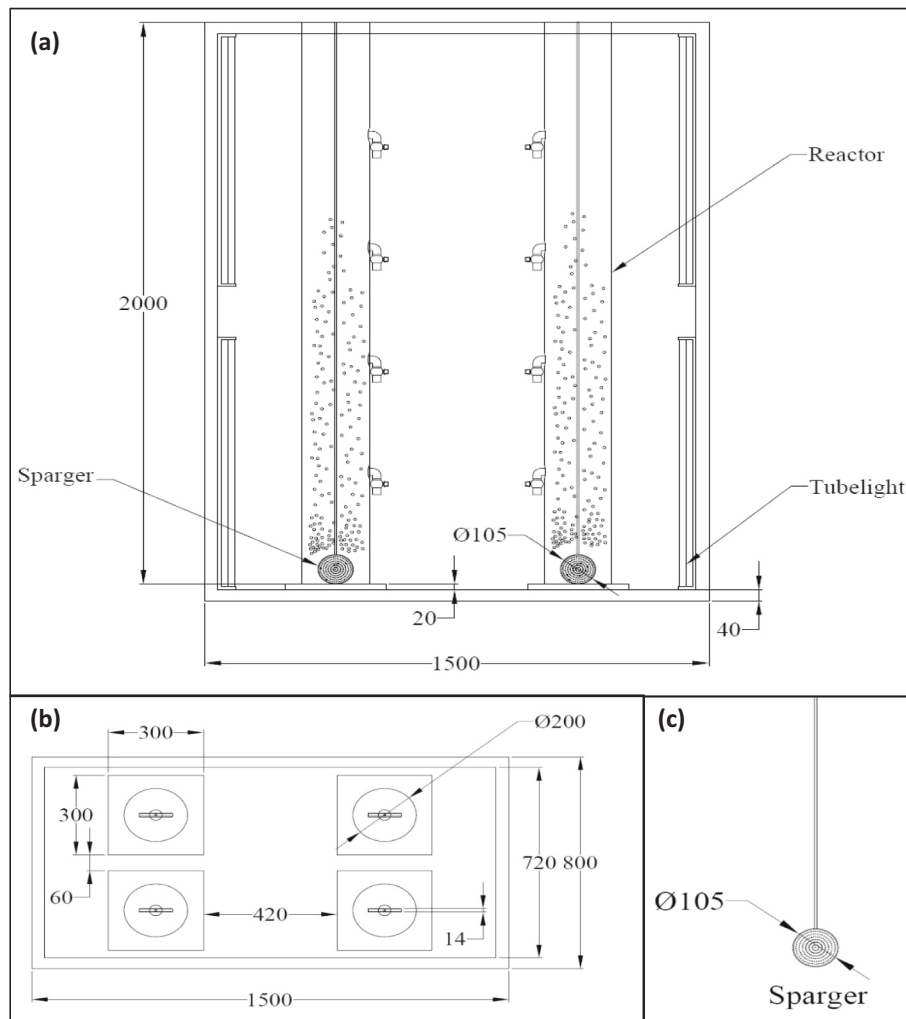


Fig. 1. Detailed drawing of Bubble column Photobioreactor (BCPBR) showing different views and components of BCPBR and drawn on AutoCAD: (a) Front view; (b) Top view; (c) Air/CO₂ sparger.

PBRs are flat panel reactors, tubular, and bubble column PBRs [3,4].

The bubble column photobioreactor stands on, with a height greater than twice their diameter, with sparger's air bubbles that suit the area and volume of the column. Air bubbles make a cloud effect on the column center that increases the path length of light. Perforated plates must be placed along the column height to improve the bubble's dispersion [5]. The growth medium is agitated by the sparger to make uniform the nutrients, pH, and gaseous transfer and prevent the sedimentation of the microalgal cells. It also increases the efficiency of microalgal cells in photosynthesis. However, the intensive mixing and high flow of bubbles caused increased bubble scattering and limited light utilization, which may destroy the microalgal cells [1].

Bose et al. (2021) used *Spirulina platensis* to produce biogas at 6.3 L/h using a lab-scale bubble column photobioreactor, and the results were scalable. The growth medium was adjusted at 9.35 pH value, at 20 °C, 40.5–67.5 $\mu\text{mol m}^{-2} \text{s}^{-1}$ light intensity, with 0.5 CO₂ liquid/gas ratio (L/G) [6]. A hanging bag photobioreactor has been performed to be operated as a bubble column PBR for the cultivation of *Chlorella sorokiniana* and *Chlamydomonas reinhardtii*. The growth parameters were adjusted at the temperature of 25 °C, light intensity reached 100 $\mu\text{mol m}^{-2} \text{s}^{-1}$, and airflow at 5 L.min⁻¹. The results showed a double-fold biomass productivity for *C. sorokiniana* and increasing of 11 % for *C. reinhardtii* [4]. Another study by Wu et al. (2021) investigated the growth of *Haematococcus pluvialis* in a microbubble-driven photobioreactor (MDPBR) with a central light source, which was between 16 and 20 mmol photo/

m²/s. They worked on decreasing the bubble size to increase CO₂ mass transfer with a 0.05 L min⁻¹ flow rate. The results showed cell growth productivity reached 0.49 per day with 5-fold higher biomass content [7]. Gu et al. (2022) investigated the ability of the microalga *Phaeodactylum tricornutum* to produce eicosapentaenoic acid (EPA) using 50 L bubble column PBR. The aeration rate was 20 L min⁻¹, with 1 % (v/v) CO₂ concentration, at light intensity started at 230 $\mu\text{mol m}^{-2} \text{s}^{-1}$ on the inoculation day, then raised to 600 $\mu\text{mol m}^{-2} \text{s}^{-1}$ after a day of cultivation and 1100 $\mu\text{mol m}^{-2} \text{s}^{-1}$ after 96 h passed. The light cycle was adjusted at 16:8 light/dark, at 22 ± 1 °C. The results recorded 1.5 g/L after the cultivation for 15 days, which contained a lot of EPA, about 5 % of cellular dry weight [8].

Microalgae cultivation in BCPBRs requires controlling some growth factors to optimize growth, such as adequate lighting, high mixing intensity, and low shear stress [9]. Bubble column PBR (BCPBR) is a suitable bioreactor for shear-sensitive microalgae cultivation because of the low shear stress produced by the aeration system, which is lower than that generated from pumping and stirring techniques [9]. Some growth factors influence microalgal growth, such as temperature, pH, carbon concentration, and growth mechanism [10,11]. In addition, BCPBRs modeling can be done by considering aeration rates, air volume fraction, bubble diameter, energy dissipation rate (EDR), shear rate, gas and liquid densities, velocity, heat and mass transfer and growth kinetics [9,10,12]. In the case of linear airflow, the air velocity can affect shear stress [13]. LED illumination has been used recently in PBRs as a light

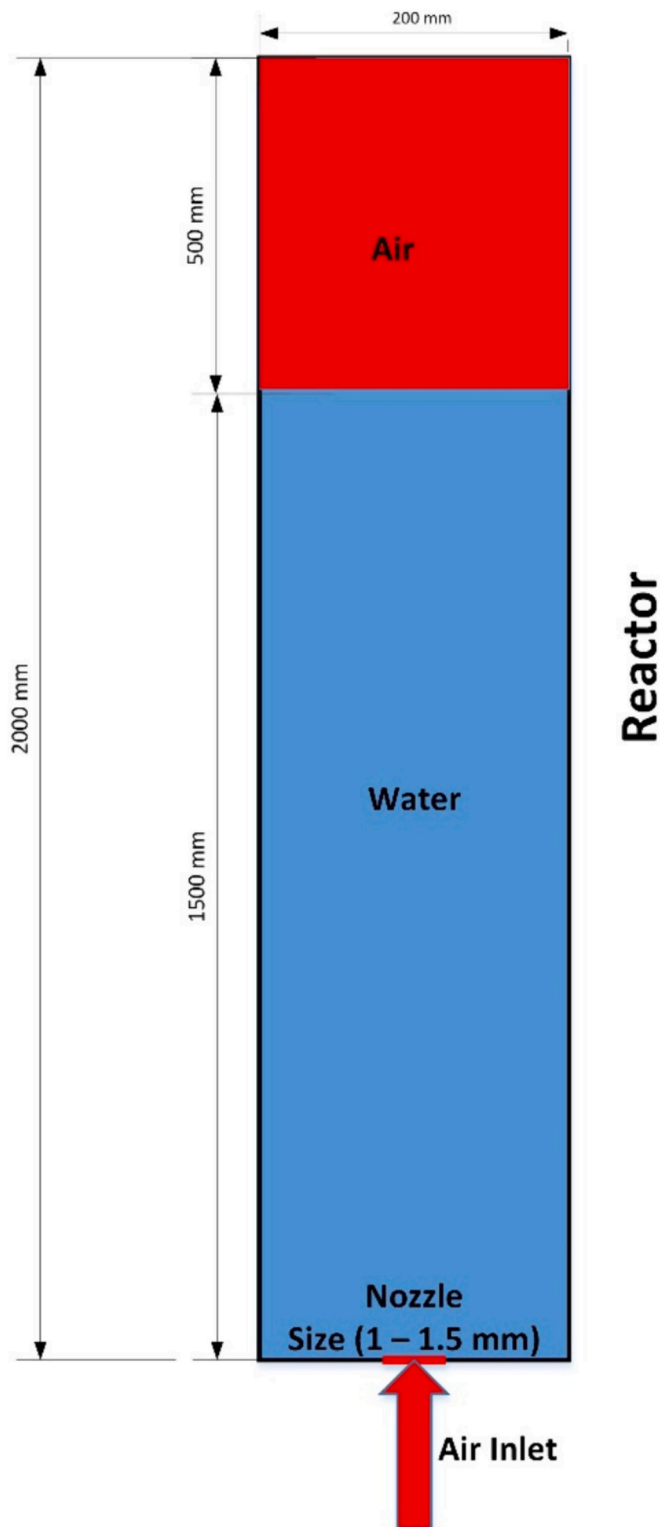


Fig. 2. The computational geometry of the individual Bubble column photobioreactor (BCPBR) indicates the two phases of water (AF6/RWW) and Air/CO₂. For simulation, the separation of phases with the dimensions is assumed.

source, which has 25 to 64 lm/W, with highly efficient energy and a strictly focused bandwidth [2]. Fouling control strategies are one of the significant factors that help clean the media and remove the microalgal films that adhere to the PBRs inner surfaces, which can cause shading and reduce photosynthesis and productivity. Talluri et al. (2020)

cultivated *Anabaena* sp. in PBRs with hydrophilic anti-fouling surfaces like glass or poly (methyl methacrylate) to overcome fouling [14]. Chanquia et al. (2022) reported the use of three physical techniques (relaxing, backwash, and nitrogen scouring) to clean the PBRs when *Chlorella vulgaris* was cultivated, the outcomes showed that backwashing, relaxation, and nitrogen clean-up both resulted in a relative reduction of fouling rate of 50 % and 60 %, respectively [2].

Computational fluid dynamics (CFD) simulation tools provide the modeling and controlling of the growth kinetics in industrial PBRs by analyzing the hydro-dynamics using a number of computer and numerical equations [5,15]. The most effective fluid dynamic factors are bubble dynamics and mixing. These parameters help in the bioreactor design, operation, cost, and efficiency [16]. CFD simulation had been used for characterizing the hydro-dynamics of an air-lift photo-bioreactor for modeling the cultivation of *Chlorella protothecoides* and *C. vulgaris* by reducing the shear stress that helped in increasing the lipid content to the produced biomass [9]. L'opez-Pacheco et al. (2021) reported using a distributor aeration tool that regulated the air micro-bubbles diameter from 0.5 mm to 1.5 mm with uniform flows, which was effective in improving microalgal growth rate by reducing the air bubbles' size in a regular flow rate [17]. CFD simulation by Ansys FLUENT (version 22.1.0) was used for modeling of microalgal in internal loop airlift PBRs with concentric tubes with consideration of cell growth kinetics, light attenuation, and solar radiation to predict the annual productivity [18]. Our prime objective was to study the turbulence generated by the bubbles, which provided the required turbulence inside the BCPBR that helped in mixing and making the culture homogeneous. We are interested in researching how nozzle size and inlet velocity affect the turbulence generated. It will help us to select the optimum nozzle size and velocity for maximum turbulence.

2. Experimental validation

2.1. Bubble column photobioreactor (BCPBR)

Researchers are more focused on growing microalgal monocultures in axenic-enclosed PBRs because these systems offer fewer or no chances for contamination in the creation of biochemical and high-value metabolites. It is a manufactured closed vessel that facilitates photosynthesis in microalgal cells using sunlight or artificial light as the energy source [19]. BCPBR was selected for this study because it possessed some advantages over other PBRs for the growth of *C. vulgaris* [20]. This closed PBR system aids in preventing contact between the enclosed microalgal cells and the environment, allowing the culturing of micro-algae species that are difficult to grow in open pond systems. This design prototype offers better control over the culture operational conditions such as temperature, pH, pCO₂, pO₂, and nutrient supply, prevents evaporation, reduces water usage, lowers CO₂ losses due to outgassing, achieves higher biomass concentration, and greater volumetric productivity. BCPBR is fitted with cool-white fluorescent lamps for illumination to provide light/dark photosynthetic cycles for high productivity of microalgal biomass. AutoCAD drawings of different views of the reactor and the photograph of the reactor in the lab are shown in Figs. 1 and suppl. Fig. S4, respectively.

The BCPBR is designed and manufactured with a circulatory fluid pump, influx regulators to regulate air/ CO₂, sampling ports, gassing tubes, and porous stones used as air/CO₂ spargers. The inner casing of the BCPBR is made up of reflective sheets for better illumination. For laboratory-scale microalgal biomass cultivation - a wide range of systems can be employed. Microalgae can be cultured in glass Erlenmeyer flasks, fermenters, panel reactors, and PBRs. In BCPBR, turbulence caused by the rising bubbles can help to redistribute the microalgae cells throughout the culture medium. This mixing helps to prevent the formation of light-limiting zones where cells at the top of the culture absorb most of the light while those below receive less. By continuously moving the cells around, turbulence ensures uniform light exposure to all cells,

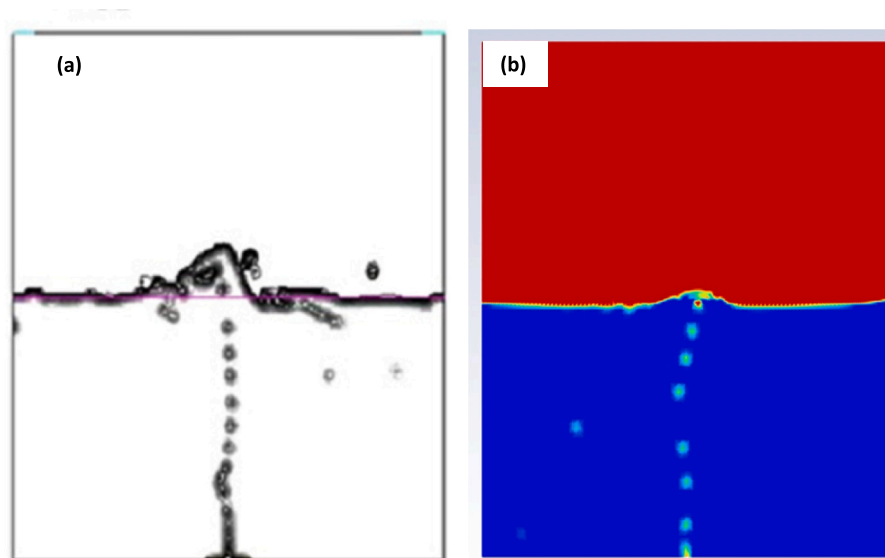


Fig. 3. Simulation of volume fraction of Air (a) Previous study; and (b) This study. The present CFD model is validated with the work by Seo et al. (2012) for the same geometry and boundary conditions. The results for the volume fraction of air are depicted in Fig. 3 (b), and it can be seen from Fig. 3 (a) that both results are extremely close to each other thereby validating our model.

enhancing photosynthesis and growth. Turbulence increases the mixing of gases and nutrients in the reactor. This improved mixing ensures that essential nutrients and gases like carbon dioxide are more evenly distributed and readily available to the microalgal (*Chlorella vulgaris*) cells. Additionally, it helps efficiently remove oxygen, which can reach inhibitory levels if accumulated. In static or less turbulent conditions, microalgae cells may settle at the bottom of the reactor, leading to suboptimal light exposure and possible cell death due to nutrient starvation. Turbulence can prevent the formation of biofilms at the walls of the reactor, which can otherwise block light and reduce the reactor's efficiency. Keeping the cells in suspension minimizes their attachment to surfaces and maintains the light transmittance inside the reactor.

2.2. Surface area/Volume ratio (S/V)

The S/V ratio of the BCPBR is defined as the ratio of the illuminable surface area of the bioreactor. The volume defines the amount of incident light transcending into the system per unit volume and the light regimen to which the microalgal cell would be vulnerable. BCPBR has an illuminated surface area of about 1.06 m^2 and a volume of 0.05 m^3 , making an S/V ratio of 21.2 m^{-1} . This will lead to higher biomass concentrations, and volumetric productivity of the microalgae culture was evident as this influenced the hydrodynamic behavior of the culture, resulting in shorter light-to-dark cycles for the PBRs [21].

2.3. Mixing

When designing an operational PBR, consideration must be given to the type and configuration of the apparatus used to circulate the microalgae suspension within the culture flux in order to increase productivity and avoid the high cost of integrating a mechanical fluid mixer, an air pump was used to drive bubbles during the aerated mixing process. Along with precisely controlling the bioreactor's circulatory flux, the coupled air pump also creates the necessary gaseous environment for the growth of microalgal cells. Thus, this prevents the settling of proliferating cells, prevents the development of temperature and pH gradients along the vertical axis of the BCPBR, and promotes effective nutrient diffusion. Additionally, it eliminates photosynthetically produced oxygen that can inhibit photosynthesis at specific levels, transports CO_2 to the membrane of the growing cells, and ensures that these growing cells experience alternating periods of light and darkness.

Although excessive mixing frequently causes the culture to foam, it is crucial to keep in mind that it can also cause cellular disruption and, eventually, cell death. Because of this, the cultivation of *C. vulgaris* within the BCPBR system required a specific level of aeration reached 2–3 lpm air + 2–3 % CO_2 v/v. The fluid mixing rate was kept at $14.4 \text{ m}^3/\text{d}$, and it was achieved intermittently by the recirculatory mechanism (bottom-top) of BCPBR.

2.4. Light intensity

Algae productivity is greatly influenced by the amount of light they receive and how effectively it is converted into biomass and lipids, all of which have an impact on their rate of growth. Algal biomass density, culture medium depth, and PBR type are the factors that impact light intensity. If the culture's cell concentration and depth are higher than average, more light is needed. Still, it should be remembered that sometimes, both direct sunlight and highly concentrated artificial light can cause photoinhibition [22]. BCPBR has a technical advantage for better light distribution among the family of PBRs. It is best to prevent the system from overheating. The optimum range of light intensity for microalgal growth and the accumulation of different value-added compounds is $65\text{--}2000 \text{ } \mu\text{mol m}^{-2} \text{ s}^{-1}$. Exclusively about *Chlorella vulgaris*, this range lies between 100 and $200 \text{ } \mu\text{mol m}^{-2} \text{ s}^{-1}$ [23]. To improve the effectiveness of photosynthesis, providing light-dark cycles is crucial. The photochemical phase needs the light cycle to make ATP and NADPH, and the biochemical phase needs the dark cycle to make the crucial biomolecules for microbial growth [24]. The light intensity maintained in the three runs was 100, 150, and $130 \text{ } \mu\text{mol m}^{-2} \text{ s}^{-1}$, respectively. Light: Dark cycles maintained for the same runs were 24:0, 12:12, and 16:8.

2.5. Sterilization of bubble column photobioreactor (BCPBR)

Generally, sterilization is obtained by the following techniques: (i) Autoclaving: Usually the most effective method. Uses high steam pressure that produces a high sterilizing temperature; (ii) Sterilization by Filtration: Good for small volumes. Uses sterile filters (pore size of $0.2 \text{ } \mu\text{m}$); and (iii) Sodium hypochlorite (Aka Bleach): Good for large volumes for aquaculture. 1 to 5 mL of bleach is added per litre, gently mixed, and allowed to stand for several hours away from direct sunlight. To neutralize the water and bleach solution, a sodium thiosulfate stock

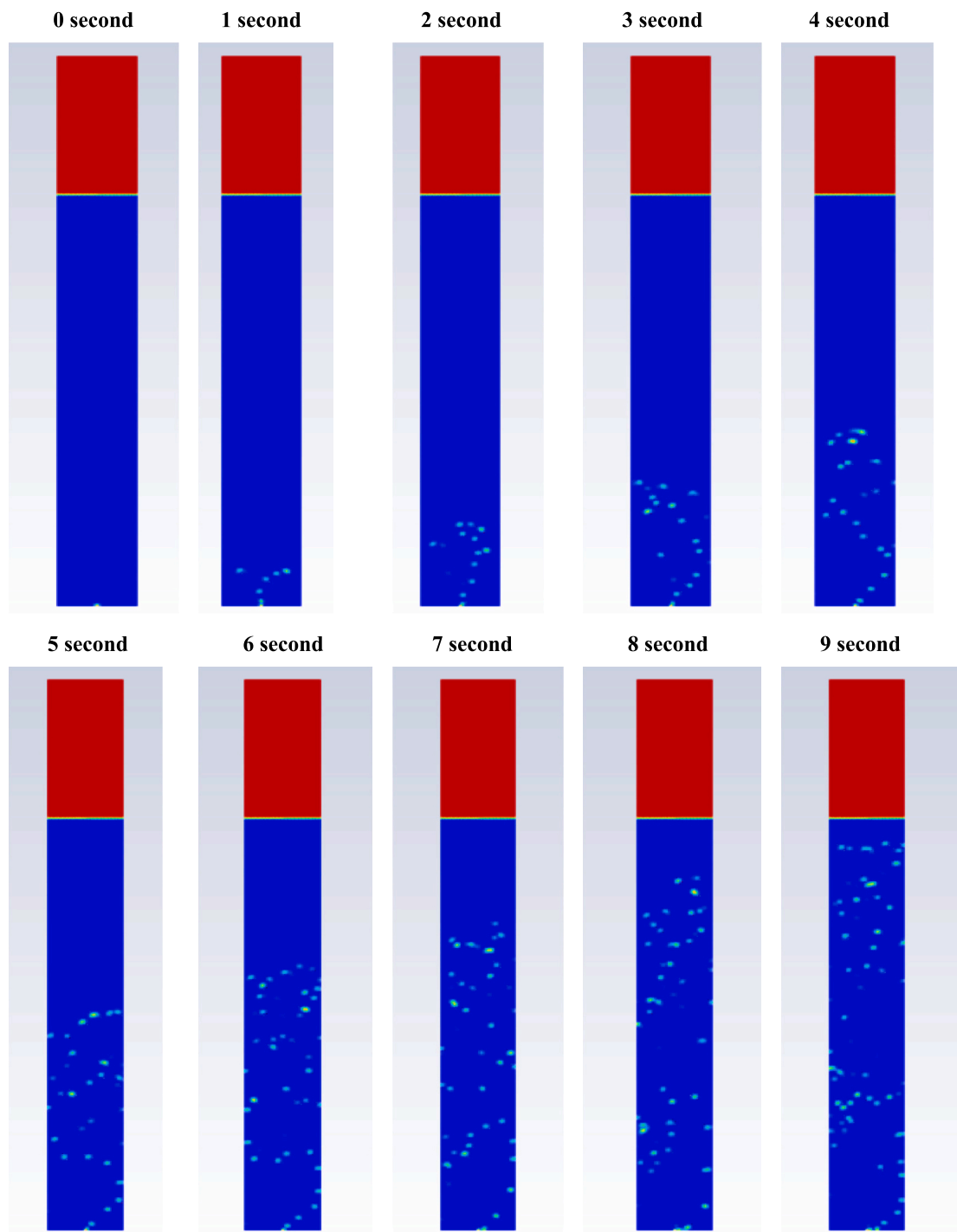


Fig. 4. Evolution of bubbles with time in the BCPBR. It shows the contours of volume fraction air injected from a 1.5 mm nozzle with a velocity of 0.3 m/s. Starting from zero time, each plot shows the contour plots after a gap of 1 s.

solution is prepared using 250 g per litre. From this, 1 mL is added to the solution for every 4 mL of bleach added.

2.6. Numerical setup for the simulation of bubbles in BCPBR

CFD analysis was conducted on the evolution of bubbles in the BCPBR. Velocity and pressure coupling are corrected using the Pressure-Implicit with Splitting of Operators (PISO) method. A second-order upwind calculation method is used to determine the flow parameters. The geo-reconstruct method is used to construct the interface. The residuals have been modified at a value of 10^{-4} . An intake with a constant

velocity is used for the water jet. A non-slip condition is applied to the wall's edges. The top of the reactor is open and designed as a pressure exit. The standard $k-\omega$ model, a semi-empirical linear eddy viscosity model, is used to describe flow turbulence. It is based on the model transport equations for the turbulent kinetic energy and its dissipation rate.

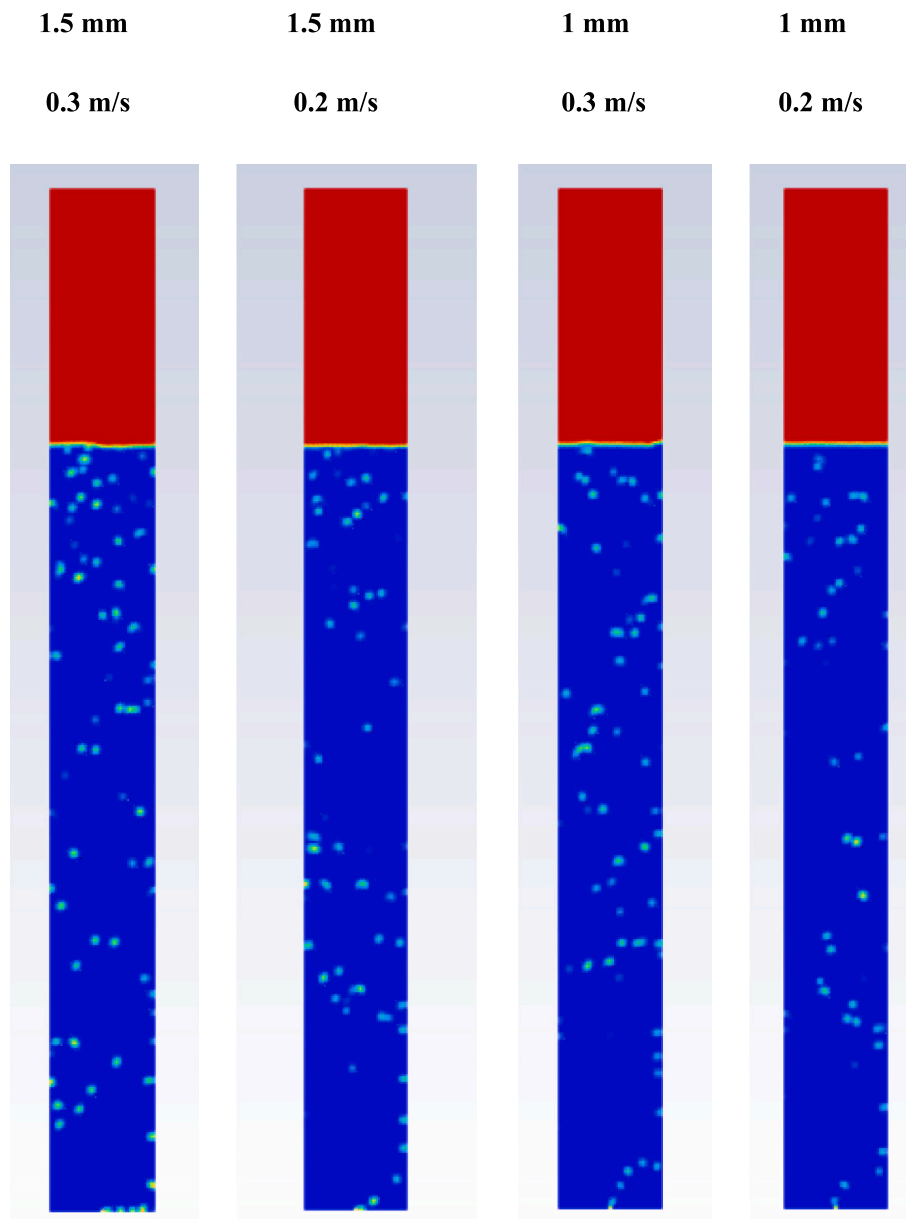


Fig. 5. Variation in the evolution of bubbles with different velocities for the set of same nozzle size at time = 10 s.

3. Results and discussion

3.1. Computational fluid dynamics (CFD) simulation of the evolution of bubbles in bubble column photobioreactor (BCPBR)

The main criteria for the selection of BCPBR for this study was the better mixing and aeration imparted by the diffusion of the bubbles. The mixing between the medium and microalgal inoculum takes place due to the turbulence created by the bubbles as they rise from the bottom to the top of the reactor system. The transfer of oxygen also takes place through the surface of the bubbles via the bubble-water interface and gas-liquid interface [25]. This feature also prevents the gas holdup inside the reactor, which is a common problem encountered in vertical reactors. CFD simulation is used to study the hydrodynamics of the culture medium inside the BCPBR. A 2D geometry of the reactor was considered, with the exact dimensions as experimental. Fig. 2 shows the computational geometry of the 2D BCPBR (1 unit). The experimental results observed during the cultivation of *C. vulgaris* with RWW in BCPBR clearly showed better mixing, high growth, and improved

treatment efficiency. The CFD will provide validation for the evolution and distribution of bubbles coming out from the sparger at the bottom and moving up in a zig-zag fashion, creating turbulence and contributing to better light distribution, CO₂ transfer, heat and mass transfer.

The total height of the individual reactor is 2000 mm, with 1500 filled with reactor fluid (AF6 or RWW) and the top 500 mm containing air. At the bottom, we have a nozzle whose diameter varies between 1 and 1.5 mm while the air velocity varies from 0.2 to 0.3 m/s. The geometry and mesh are generated in the Design Modeler, while Ansys Fluent (2022-R1 students' version) is used for flow simulation.

To compare BCPBR performance quantitatively, the efficient mixing that was expected in BCPBR and discussed in the literature was applied to the CFD model. The experimental results observed during the cultivation of *Chlorella vulgaris* with RWW in BCPBR clearly showed better mixing, high growth, and improved treatment efficiency. *Chlorella vulgaris* in the BCPBR was cultivated using AF6 as a control and RWW (Figs. S1 & S2). Growth was observed by measuring the optical density at 750 nm and dry cell weight every alternate day, followed by the determination of specific growth rate (day⁻¹) and biomass productivity

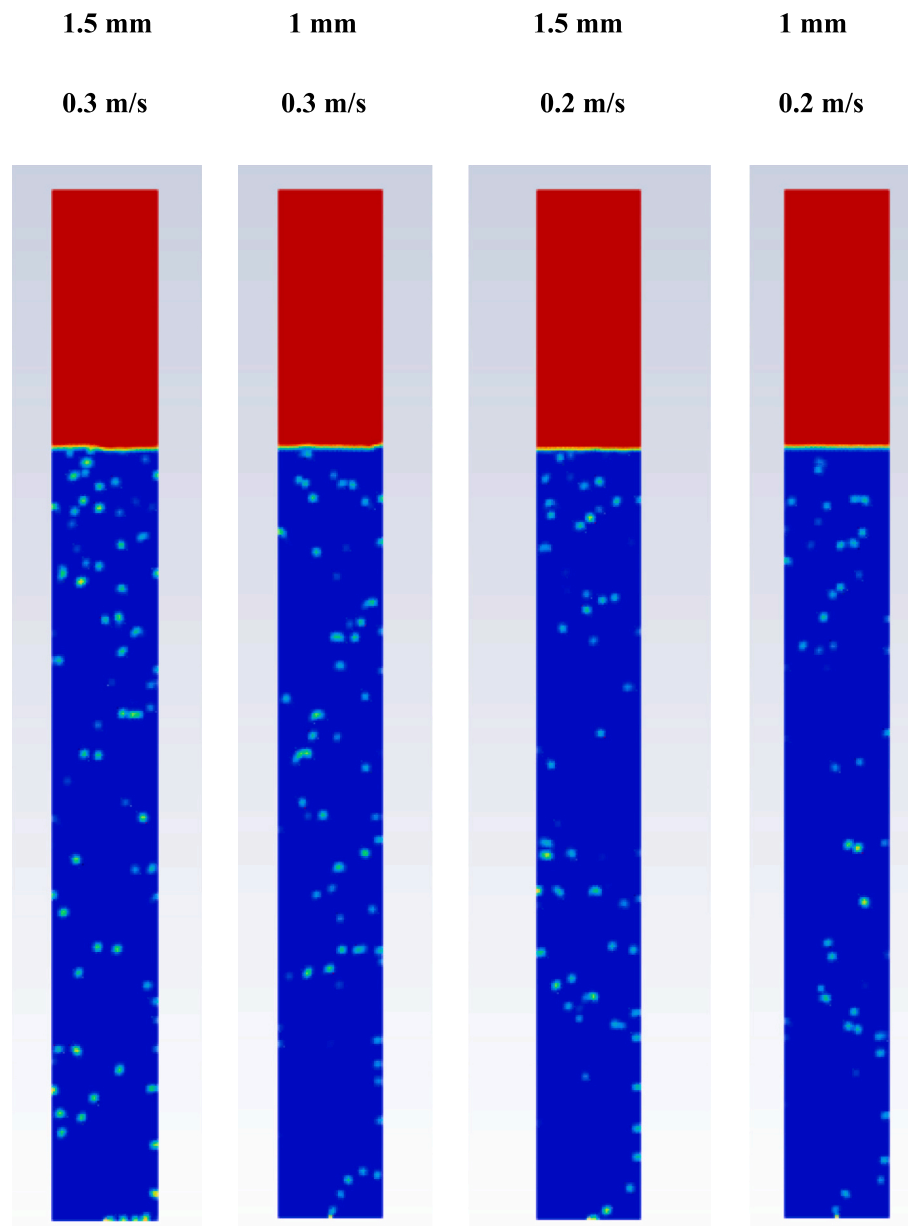


Fig. 6. Variation in the evolution of bubbles with different nozzle sizes for the set of same velocities at time = 10 s.

(mg/L/d). It was observed that the growth of *Chlorella vulgaris* showed resilience and acclimatization with the RWW.

The dry cell weight and biomass productivity are almost the same for control and RWW. The cultivation in BCPBR is carried out using light and dark cycles because microalgal photosynthesis includes light and dark reactions (Calvin cycle). Light reaction cycles occur in the thylakoids, converting light energy to chemical energy, while in the dark cycle, the active chemical energy is transformed into stable chemical energy. This shows the compositional richness (adequate concentrations of C, N, and P) of RWW and the efficacy of selected BCPBR, which supported the fast growth of *Chlorella vulgaris*. The dense culture with a dark green color prevailing inside the BCPBR indicates healthy growth and high biomass production. The trend of removal of COD, BOD, and FOG with time and the influent-effluent percentage of removal are shown in Fig. S3, respectively. It was found that when *Chlorella vulgaris* is cultivated with RWW, the maximum removal achieved was 94 %, 97 %, and 86 % for COD, BOD, and FOG, respectively. That is a clear indication of homogenous mixing between RWW and the inoculum of *Chlorella vulgaris*.

Therefore, CFD simulation in this study will contribute to the following two aspects: (i) the volume fraction contours and velocity contours are going to validate the experimental study, and (ii) the outcomes might serve as a method for developing better modules of BCPBRs for microalgae culture.

3.2. Multiphase model

In the PBR, air bubbles are used to mix the culture medium internally and supply carbon dioxide. There were many multiphase models available in fluent. As recommended by Seo et al. (2012), in the current work, the Volume of Fluid (VOF) Model was utilized [26]. For each phase, governing equations must be solved, and the VOF technique condenses the mass and momentum equations into a single set of equations. The VOF is used with the level-set approach to improve interface tracking, which was created specifically for situations without transferring any mass between phases. The smooth and continuous nature of the level-set function allows for accurate computation of its spatial gradients.

In this approach, the formulas for the conservation of mass and

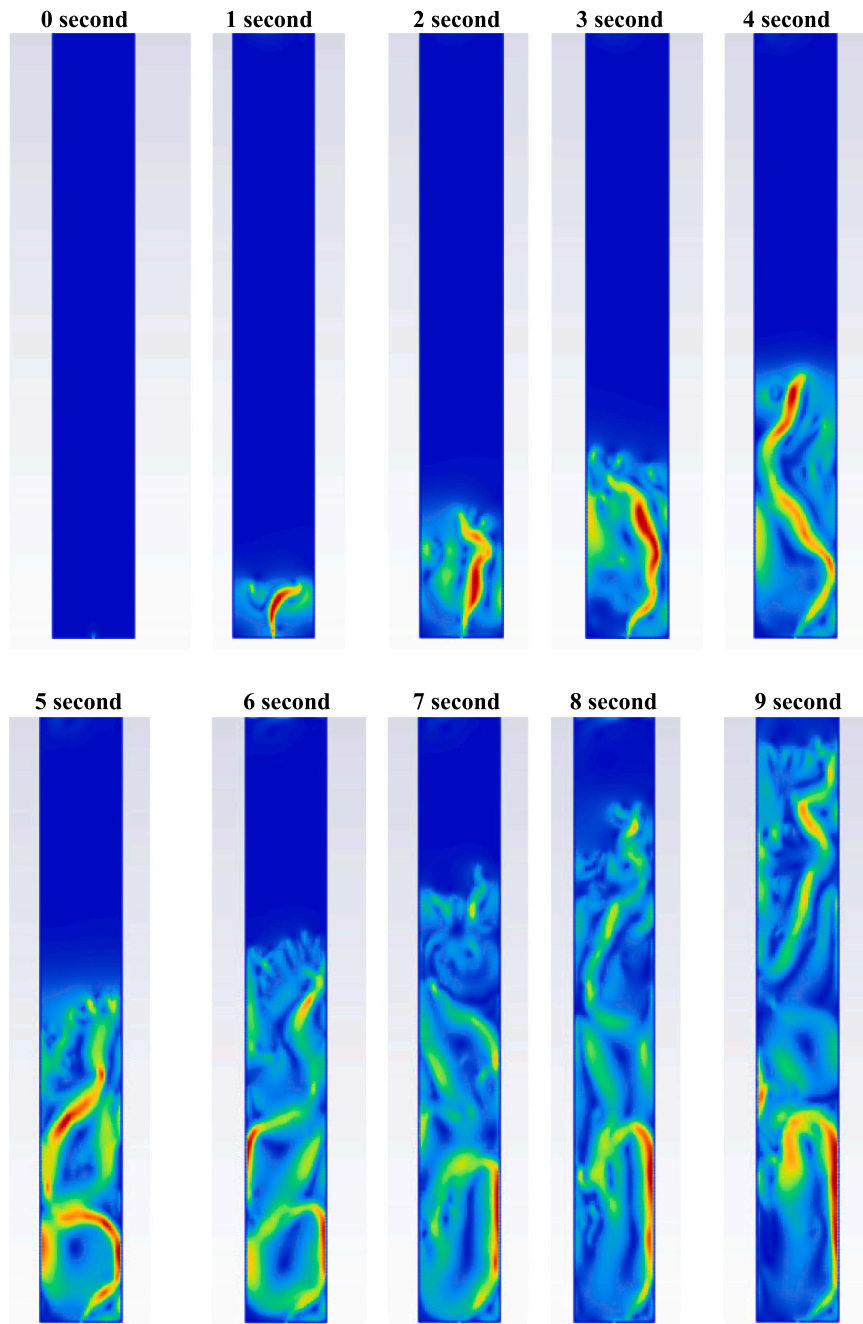


Fig. 7. Evolution of air velocity contours with time for a specific nozzle size of 1.5 mm at a velocity of 0.3 m/s. Here, we also start from zero time, and each plot shows the contour plots after a gap of 1 s.

momentum are Eqs. 1 and 2, respectively:

$$\nabla \cdot (\vec{V}) = 0 \quad (1)$$

$$\frac{\partial(\rho \vec{V})}{\partial t} + \nabla \cdot (\rho \vec{V} \vec{V}) = -\nabla p + \nabla \cdot [\mu (\nabla \vec{V} + \nabla \cdot \vec{V} \vec{T})] + \rho \vec{g} - \sigma K \delta(\varphi) \nabla(\varphi) \quad (2)$$

where $\vec{V}T$ is the stress tensor, φ is the level-set function, σ is the surface tension coefficient, and K is the curvature.

The level-set function, which is a function of the final term on the right side of the momentum equation, is used to calculate the phase interaction between the fluids. Since the interface is the zero level-set in a two-phase flow system, it is defined as a signed distance to the inter-

face and can be represented as $\Gamma = \{x | \varphi(x, t) = 0\}$; where φ can be defined as Eq. 3.

$$\varphi = \begin{cases} +|dI| & \text{if } x \in \text{the primary phase} \\ 0 & \text{if } x \in \Gamma \\ -|dI| & \text{if } x \in \text{the secondary phase} \end{cases} \quad (3)$$

Where dI is the distance from the interface.

The interface's normal (nI) and curvature (K) are necessary in order to compute the surface tension force and estimated as Eqs. 4 and 5, respectively:

$$\vec{n}I = \frac{\nabla \varphi}{|\nabla \varphi|} \bigg|_{\varphi=0} \quad (4)$$

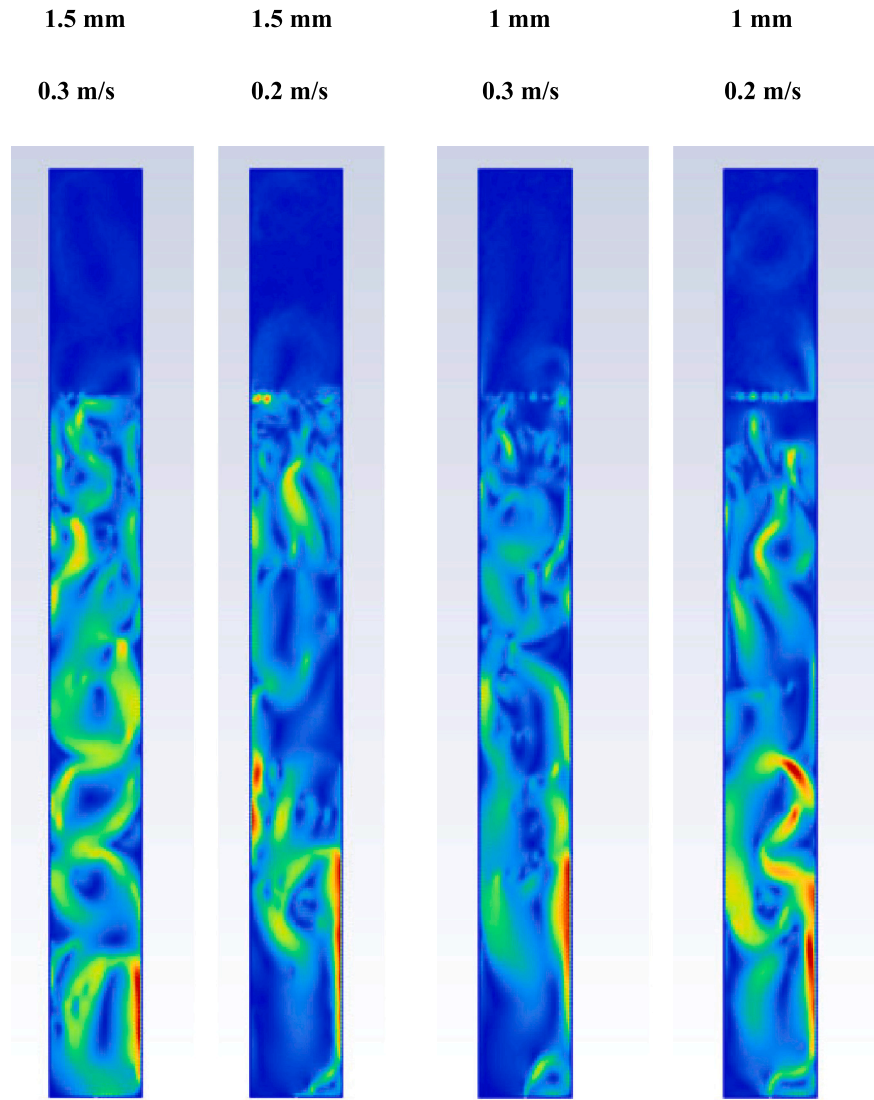


Fig. 8. Variation in velocity contours by keeping the uniform nozzle sizes- varying the velocities at time = 10 s.

$$K = \nabla \cdot \frac{\nabla \varphi}{|\nabla \varphi|} \bigg|_{\varphi=0} \quad (5)$$

Eq. 6 is then used to represent the level-set function's transport equation, and Eq. 7 is used to represent the solution of Eq. 7 for the secondary phase's conservation of the volume fraction.

$$\frac{\partial \varphi}{\partial t} + \vec{V} \cdot \nabla \varphi = 0 \quad (6)$$

$$\frac{\partial \alpha_p}{\partial t} + \vec{V} \cdot \nabla \alpha_p = 0 \quad (7)$$

After that, Eq. 8 is used to calculate the volume fraction equation for the primary phase. The properties that appear in the transport equations depend on whether the component phases are present. Eqs. 9 and 10 are then used to calculate the mixture's density and viscosity.

$$\alpha_q = 1 - \alpha_p \quad (8)$$

$$\rho = \alpha_q \rho_q + \alpha_p \rho_p \quad (9)$$

$$\mu = \alpha_q \mu_q + \alpha_p \mu_p \quad (10)$$

Where α is the volume fraction, subscripts p and q denote phases p and q, ρ is the density of the mixture; subscripts p and q denote phases p and

phase q (kg/m^3), and μ is the viscosity of the mixture; subscripts p and q denote phases p and phase q (kg/m.s).

3.3. Mesh independence and model validation

We have considered three mesh sizes with 6317, 10,869, and 15,572 nodes in the present analysis. The maximum velocity magnitude is found to be 0.092 m/s, 0.085 m/s, and 0.084 m/s. The variation in maximum velocity between 10,869 nodes and 15,572 nodes is only 1.19 %. Thus, we have used 10,869 nodes in the present analysis. The present CFD model is validated with the work carried out by Seo et al. (2012) for the same geometry and boundary conditions [26]. The results for the volume fraction of air are depicted in Fig. 3 (b), and it can be clearly seen from Fig. 3 (a) that both results are incredibly close to each other, which validates our model. Our prime objective was to study the turbulence generated by the bubbles, which provided the required turbulence inside the BCPBR that helped in mixing and making the culture homogeneous. The impact of nozzle size and inlet velocity on the turbulence produced is something we want to research. It will help us select the optimum nozzle size and velocity for maximum turbulence. The simulations were performed for nozzle size of 1–1.5 mm, and velocity of inlet air ranged between 0.2 and 0.3 m/s. The results are shown in terms of volume fraction and velocity contours that assist in understanding the

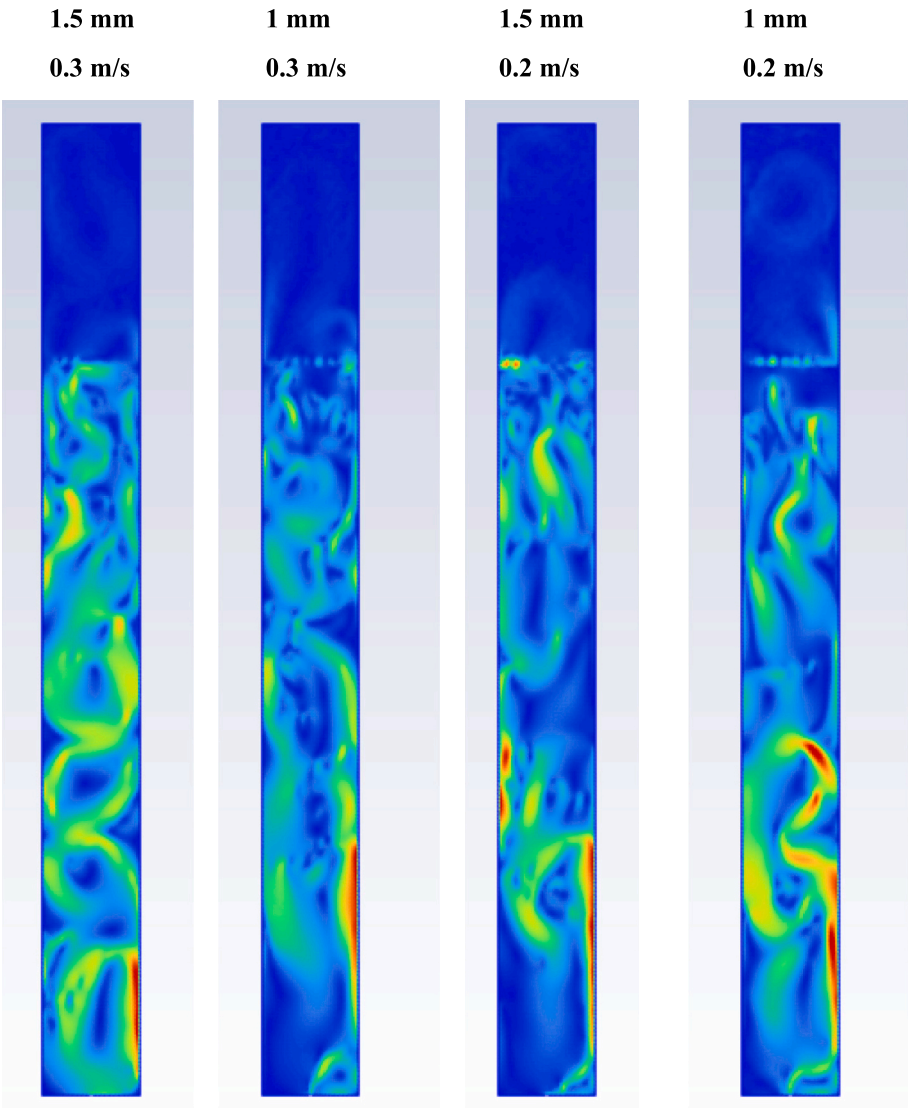


Fig. 9. Variation in velocity contours by keeping same velocities but varying the nozzle sizes at time = 10 s.

Table 1

Turbulence parameters for different simulation cases.

Simulation case nozzle size (velocity)	Max. turbulent kinetic energy m ² /s ²	Max. turbulent intensity %	Max. Turbulent viscosity Kg/m/s
1.5 mm (0.3 m/s)	0.0237	12.582	0.749
1.5 mm (0.2 m/s)	0.0196	11.452	0.6574
1.0 mm (0.3 m/s)	0.0255	13.038	0.7684
1.0 mm (0.2 m/s)	0.0147	9.906	0.6432

distribution of bubbles and their turbulence in the BCPBR.

3.4. Volume fraction contours with time

Fig. 4 shows the contours of volume fraction air injected from a nozzle of size 1.5 mm with a velocity of 0.3 m/s. Initially, it was assumed that out of the 2 m height of the reactor, 1.5 m is filled with reactor fluid while the top 0.5 m is filled with air. In the contour plot, the red color on the top shows that it is filled with air, while the blue color shows the reactor fluid, which is water in the present case. Starting from zero time, each plot shows the contour plots after a gap of 1 s. As can be seen from the air bubbles originating from the center, they initially move upwards

and then drift slightly towards the right; after that, they move up, inclining towards the right side of BCPBR. During the experimental growth, biofouling was observed at the left side of BCPBR, which signifies that the mixing required was not attained. The bubbles move more randomly, thereby generating the turbulence that is required for better mixing of reactor fluid. It is also observed that as bubbles move upwards, they cover almost the entire width of the reactor. It can also be observed that the bubbles reach almost half of the height of the reactor in five seconds, while they reach the top in about ten seconds.

A similar pattern is observed for all the nozzle sizes and velocities considered in the present analysis. Fig. 5 shows the variation in bubble distribution with velocities for the same nozzle size. At higher velocities, a more significant number of bubbles existed in the reactor. Higher turbulence can be seen as expected because of a more significant number of bubbles. A similar pattern is observed for other nozzle sizes. Fig. 6 shows the variation in bubble distribution with nozzle sizes for the same velocities. For a larger size nozzle, more bubbles are present in the nozzle. Some coalescence is also observed at large nozzle sizes. Further, it is also witnessed that for smaller nozzle sizes, the initial movement of bubbles is in the upward direction. In contrast, large nozzle sizes are in the horizontal direction. This may be because of larger buoyancy forces in smaller bubbles.

3.5. Velocity contours with time

Fig. 7 shows the evolution of air velocity contours with time for a specific nozzle size of 1.5 mm at a velocity of 0.3 m/s. Here, we also start from zero time, and each plot shows the contour plots after a gap of 1 s. The contour plots of air velocity move in a zig-zag way, thus creating turbulence in the reactor. This randomness of moving bubbles generates the turbulence required for better mixing reactor fluid. Recirculation zones are also observed that engulf the entire width of the reactor. It can also be observed that the contours reach almost half the reactor's height in five seconds and up to the top in about ten seconds. Similar patterns are observed for all the nozzle sizes and velocities used in the present analysis.

Fig. 8 shows the contour plots of velocities for the same nozzle size. At higher velocities, more recirculation zones are observed in the reactor. Thus, higher velocities result in higher turbulence for the exact size of the nozzle. A similar pattern is observed for other nozzle sizes. Fig. 9 shows the contour plots of velocity with varied nozzle sizes. At a velocity of 0.3 m/s, more recirculation zones are observed for a nozzle size of 1.5 mm compared to a nozzle size of 1 mm. It is also witnessed that for smaller nozzle sizes, the difference in contour plots is not that much, indicating almost a similar turbulence level for both nozzle sizes.

3.6. Turbulence in the bubble column photobioreactor (BCPBR)

Turbulence is one of the most essential factors in BCPBR, as the required mixing in the reactor is achieved by the turbulent kinetic energy of the bubbles. Table 1 summarizes the turbulence parameters for the simulations performed during this study. It is clearly evident that turbulence parameters increase with an increase in inlet velocity for a fixed nozzle size and an increase in nozzle size for a fixed velocity. That occurs when the nozzle diameter is maintained at 1.5 mm while the velocity is increased from 0.2 to 0.3 m/s and the turbulent intensity increases from 11 to 12 %. Similarly, when the nozzle diameter is increased from 1 mm to 1.5 mm by keeping the velocity uniform at 0.2 m/s, the turbulent intensity increases from 10 to 11 %. Overall, while the effects of a 1 % increase in turbulence intensity are likely to be subtle, they could still contribute to more efficient treatment and increased productivity, especially in BCPBR, where even minor optimizations can lead to improved outcomes. Continuous monitoring and adjustment based on observed effects would be essential to fully leverage such a minor increase in turbulence intensity.

4. Conclusion

- CFD analysis for the distribution of the bubbles in the BCPBR was carried out by plotting the volume fraction and velocity contours with respect to the nozzle size (1.0/1.5 mm) and up flow velocity (0.2/0.3 m/s) of the bubbles.
- Turbulence parameters increase with an increase in inlet velocity for a fixed nozzle size and an increase in nozzle size for a fixed velocity. Turbulence increases with an increase in inlet velocity for a fixed nozzle size and an increase in nozzle size for a fixed velocity.
- That is, when the velocity is raised from 0.2 to 0.3 m/s at the same nozzle diameter of 1.5 mm, the turbulent intensity increases from 11 to 12 %. At the same time, when the nozzle diameter is increased from 1 mm to 1.5 mm by keeping the velocity uniform at 0.2 m/s, the turbulent intensity increases from 10 to 11 %.
- The increase in turbulence inside the BCPBR leads to improved mixing-making the culture homogenous, better light distribution-improving the photosynthetic reactions, and a high growth rate-increasing the productivity, thereby leading towards efficient treatment of RWW using *C. vulgaris*.

Ethical approval

Not applicable.

CRediT authorship contribution statement

Imran Ahmad: Writing – original draft, Conceptualization, Methodology, Validation. **Shahid Husain:** Writing – review & editing, Software, Formal analysis. **Lamia H.S. Hassan:** Writing – review & editing, Visualization, Project administration. **Nadeem A Khan:** Writing – review & editing, Conceptualization, Validation. **Sasan Zahmatkesh:** Supervision, Project administration, Funding acquisition. **Bing Jie-Ni:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Availability of data and materials

The data supporting this study's findings are available from Imran Ahmad. Still, restrictions apply to the availability of these data, which were used under license for the current research and are not publicly available. However, data are available from the authors upon reasonable request and with the permission of Imran Ahmad.

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Animal research

Not applicable.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.algal.2024.103725>.

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